India’s growth story: Is it really jobless?

Going beyond the obvious with a focus on women and youth

A White Paper

November 2025

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TABLE OF CONTENTS

ABSTRACT

This white paper investigates the relationship between economic growth and employment generation in India, emphasizing the critical role of inclusive growth that actively integrates women and youth into the nation's development process. Despite India's consistent GDP expansion over the past decade, the pace of job creation has lagged, resulting in periods of jobless growth. The paper analyses the structural and institutional causes of this phenomenon, including increasing capital intensity, the dominance of service-led growth, skill mismatches, and regulatory barriers that hinder labor absorption. It further explores how improving the ease of doing business can catalyse entrepreneurship, investment, and formal sector employment. Central to this study is the recognition that sustainable economic progress requires inclusive participation. Expanding opportunities for women and youth enhances productivity, reduces inequality, and strengthens social resilience. Using comparative international data from 2017 to 2024 and examining regional patterns across Indian states, the paper underscores the need for policy reforms that foster labor intensive growth, promote gender equity, and empower young workers, thereby ensuring that India's economic growth becomes both inclusive and transformative.

INTRODUCTION

Economic growth is one of the most fundamental measures of a nation's progress. It is typically assessed through an increase in per capita national income, which reflects an expansion in productivity, investment, and the overall well-being of citizens. A growing economy stimulates demand, enhances production of goods and services, and attracts both domestic and foreign investment. It also strengthens public finances by increasing tax revenues, which governments can reinvest in essential public services such as education, healthcare, welfare programmes, and infrastructure (World Bank, 2023). These reinvestments, in turn, contribute to improved human capital, better living standards, and greater social welfare.

For India, maintaining high and inclusive economic growth is vital to achieving its developmental goals. Rapid growth supports poverty reduction, job creation, and the expansion of infrastructure in transport, power, and digital connectivity. Furthermore, a robust economy strengthens India's global competitiveness, improves its standing in international trade, and enhances its participation in the technology and innovation sectors. Yet, despite sustained economic expansion in recent decades, the challenge of ensuring that growth translates into adequate employment opportunities persists. This disconnect has led to debates about the nature and inclusivity of India's growth process.

Economic Growth and Employment Generation

Economic expansion is often associated with increased private investment, technological advancement, and infrastructure development, all of which can generate employment opportunities. However, the degree to which growth translates into job creation depends on several factors, including the composition of growth across sectors, the technological intensity of production, and the skill level of the workforce.

In India, economic growth over the past two decades has been relatively high but uneven in its impact on employment. Certain sectors, particularly services and capital-intensive industries, have contributed significantly to GDP but have not generated sufficient employment opportunities. As a result, a large segment of the workforce continues to be engaged in low-productivity or informal sector activities. This situation has given rise to what economists describe as **jobless growth**, wherein the economy grows without a proportional increase in labor absorption

Understanding Jobless Growth

Jobless growth is a multifaceted phenomenon resulting from structural, technological, and institutional factors. One major cause is the increasing **capital intensity** of production. A second reason is the **sectoral composition** of growth. In India, the service sector, including finance, information technology, and telecommunications, has driven much of the economic expansion. Consequently, growth in high-productivity service sectors does not necessarily translate into large-scale employment generation. Third, **rigidities in the labor market** and an unfavourable business environment have constrained formal job creation. Furthermore, inadequate access to affordable credit, slow contract enforcement, and logistical bottlenecks increase the cost of doing business.

The Role of Women and Youth in Inclusive Growth

The inclusion of women and youth in economic development is essential for achieving equitable and sustainable growth. Together, these groups represent the most dynamic yet underutilised sections of India's population. Their participation is not only a social justice imperative but also a strategic economic necessity.

When women are educated and gain access to employment opportunities, they contribute significantly to labor productivity, household incomes, and community welfare. Empirical research shows that gender equality in the labor market is associated with higher GDP growth and better human development outcomes (UNDP, 2023). However, India's female labor force participation rate remains among the lowest globally. Barriers such as lack of childcare facilities, gender-based wage gaps, safety concerns, and cultural norms continue to limit women's economic engagement.

India possesses one of the youngest populations in the world, with a median age of around 28 years (Census of India, 2021). This demographic structure presents both opportunities and challenges If adequate jobs and skills are not provided, this demographic potential may transform into a demographic burden, resulting in unemployment and social discontent.

Investment in quality education, vocational training, digital literacy, and entrepreneurship is essential for enabling youth to participate productively in the economy. Aligning skill development programmes with industry requirements and promoting start-up ecosystems can further enhance youth employability and innovation capacity.

Scope of the Paper

Economic growth alone does not guarantee employment generation or social inclusion. The quality, composition, and inclusiveness of growth are equally important. India's recent experience underscores the need for a growth model that prioritises job creation, skill development, and the participation of underrepresented groups such as women and youth.

This white paper analyses the structural transformation of India's economy from agriculture to non-agricultural sectors and assesses whether this transition has translated into employment generation during the period **2017–18 to 2023–24**. It examines India's performance in comparison with 217 other countries and analyses variations across states to identify regional disparities in employment outcomes.

LITERATURE REVIEW

Global Experiences in Employment and Growth

The recent literature emphasizes that India's labor market policies overly favour capital- and skill-intensive sectors, sidelining rural areas and MSMEs, leading to persistent jobless growth and informality. Geeta Verma (2023) argues for reforms, including more inclusive labor laws, expanded social protection, and schemes like MGNREGA to foster rural employment. In contrast, countries like South Africa successfully integrate informal workers using social grants and vocational training. Supporting these points, the 2023 State of Working India report shows over 80% of workers lack formal contracts, with limited inclusion for women and marginalized groups. Globally, Germany and Japan demonstrate that dual vocational training and strong labor protections yield robust formal labor markets. The World Bank (2024) sees fragmented policies as a major obstacle, recommending coordinated, gender- and region-sensitive policies inspired by practices in Turkey and Brazil that improve formalization and job quality.

Overview of growth-employment dynamics

The literature reflects a divergence in opinion about whether there is jobless growth or not, leaving the debate open. The data from CMIE's Consumer Pyramids Household Survey indicates that the unemployment rate in India rose to 9.2 percent in June 2024. Lack of job opportunities is further compounded by job losses on account of technological advancement, automation, consolidation and rightsizing (Jha & Mohapatra, 2020) (Madhavan N, 2018). On the other hand, (Abhi Chhabra, 2025) notes that while 2000-2021 was largely jobless growth, the recent period (2021-2024) shows improvement as increasing government capital expenditure, and the evolving start-up ecosystem are creating job opportunities, especially post-COVID. If growth were truly jobless, then one can expect weak consumption growth, but as per Household Consumption Expenditure Survey, consumption has continued to grow, which suggests employment must also be rising because consumption is tied to income which is tied to employment (Nilanjan Ghosh & Bardhan, 2024). The Government of India (PIB, 2024) reports that around 170 million jobs were created between 2016-17 and 2022-23, indicating that unemployment rates are falling.

Structural Transformation and Employment Trajectory

Unlike other major growing economies, India witnessed an unusual structural shift by leapfrogging from agriculture to services (Aggarwal, 2018) (Krishna et al., 2017) - sluggish agriculture-led growth until 1992, diversification and formal job creation post-liberalization, and service-led but fragmented expansion during the 2010s (sen et al., 2017). This distorted transformation led to declining levels of employment elasticity from the 1990s to the 2010s (Aggarwal, 2018) (Krishna et al., 2017), yielding an incredibly low elasticity of 0.04 between 2011-2019 while GDP grew at 6.67% CAGR (Basole, 2022). Though the Kuznets Process occurred largely as expected (Basole, 2022) in moving workers out of low-productivity agriculture but unsustainable construction sector emerged as the largest employment absorber (Krishna et al., 2017), contributing 10.6% of the workforce by 2012 (Aggarwal, 2018), thereby rendering the structural reallocation effects on employment negligible. Moreover, India's growth was largely productivity-driven, with major contributions from high-productivity services (70% of productivity gains between 1994–2002) and manufacturing (31% during 2003–2011), bypassing employment creation in these sustainable sectors (Krishna et al., 2017). In recent years, however, employment elasticity for the period 2017–23 was estimated at 1.11 (Nilanjan Ghosh & Bardhan, 2024) as the Worker Population Ratio (WPR) increased by almost 26% from 2017 to 2023 and the unemployment rate declined sharply from 6.0% to 3.2% (PIB, 2025). The formal economy nearly doubled employment since 2005 and raised wages as its GDP share reached 56% by 2022, yet only ~15% of workers are formal, with agriculture still dominant (Mundra, 2024). Despite high GDP growth, employment generation has lagged, highlighting weak structural transformation with limited creation of formal jobs (Zico Dasgupta & Amit Basole, 2023), indicating the need to effectively leverage India's demographic dividend by 2050 (Alonso & MacDonald, 2024).

Untapped Potential of Women and Youth

India's demographic dividend is undermined by persistently low and precarious employment, especially for women and youth. Women's economic empowerment is essential for inclusive growth, yet structural barriers such as unpaid care burdens (Evans 2023), limited access to finance and digital resources, and safety concerns keep most women in informal or agricultural work (ADBI 2023). Women's participation follows a U-shaped path shaped by education and workplace flexibility, while youth face high joblessness, informality, and skill gaps—underscoring the urgency of structural reforms and inclusive labor market strategies (Evans, 2023). This is reflected in India's stagnant female labor force participation rate (FLFPR), far below men's, with temporary increases during crises reflecting "distress employment" (Dhamija and Chawla 2023). Educated youth also face high unemployment due to skill and job mismatches and weak formal job creation, reinforcing what has been described as jobless growth (ILO 2022). FLFPR is the strongest predictor of youth unemployment, surpassing GDP growth, which underscores inclusion as the central challenge (Mundra 2024). NITI Aayog's report, "From Borrowers to Builders: Women's Role in India's Financial Growth Story," provides a data-backed analysis of the growing financial inclusion of women, highlighting their significant contributions as entrepreneurs and participants in the financial sector. The report argues that policies promoting women's entrepreneurship and access to credit are shifting their role from passive beneficiaries to active drivers of India's economic growth

Role of the State in advancing Inclusive Growth

The central government's Gender Budgeting Knowledge Hub by providing gender‑disaggregated data, best practices, and tools for gender‑responsive planning and budgeting, helps governments integrate the goal of inclusive growth and empowerment directly into policy design. Governments in India, have implemented several initiatives in tandem with various SDGs. SDG 1: no poverty, SDG 5: gender equality, SDG 8: decent work and employment and SDG 10: reducing inequalities. The Pradhan Mantri Jan Dhan Yojana (PMJDY) has promoted financial inclusion (Planning Commission 2015 Report), Kerala Academy for Skills Excellence (KASE) has trained thousands of youth, facilitating job placements in both domestic and international markets (Rao & Srinivas, 2018). "Maharashtra Employment Guarantee Scheme" targets rural employment generation , with a specific focus on women and marginalized communities (Deshmukh, 2020). Similarly, Uttar Pradesh, with its "Ambedkar Youth Employment Scheme" and Rajasthan's Vishwakarma Yuva Udyami Protsahan Yojana provide job training, entrepreneurship support, career counselling, and placement support for marginalized youth from SC and ST communities (Chandra & Pandey, 2020). Whereas, Uttarakhand's emphasis on tourism, agro-processing, and hydropower has led to employment growth in these sectors (Yadav, 2021).

In conclusion, the provided literature reveals a complex and contested narrative around India's "jobless growth." While historical data pointed to a low employment elasticity and a structural shift that favoured low-productivity sectors like construction, recent trends suggest a more optimistic picture with signs of employment generation and a declining unemployment rate. However, this positive trajectory is tempered by persistent structural issues, particularly the vulnerability of women and youth who face significant barriers to formal, sustainable employment. The existence of government schemes aimed at financial inclusion, skill development, and targeted employment for marginalized groups indicates a policy recognition of these challenges. Taken together, the literature suggests creating high-quality, sustainable jobs in productive sectors and ensuring that the benefits of growth are equitably distributed across all demographics to truly harness India's demographic dividend.

KEY PERFORMANCE INDICATORS (KPIs)

A comprehensive evaluation of employment performance requires identifying indicators that capture both quantitative and qualitative aspects of labor market outcomes. The following framework outlines five categories of Key Performance Indicators (KPIs):

1. **Job Creation**
   This dimension assesses the economy's capacity to generate employment in response to growth and structural change. Indicators such as the *Employment Growth Rate* and *Elasticity of Employment* measure job creation relative to economic expansion, while *Workforce Shifts* capture movements of labor from low to high productivity sectors.

2. **Quality of Jobs**
   Job quality reflects the degree of formalization and security within the labor force. Indicators such as the *Regular Wage or Salaried Share* and *Social Security Coverage* assess job stability, while the *Informal Employment Share* highlights vulnerable work. The *Gender Pay Gap* measures wage equity, reflecting inclusion and fairness in workplaces.

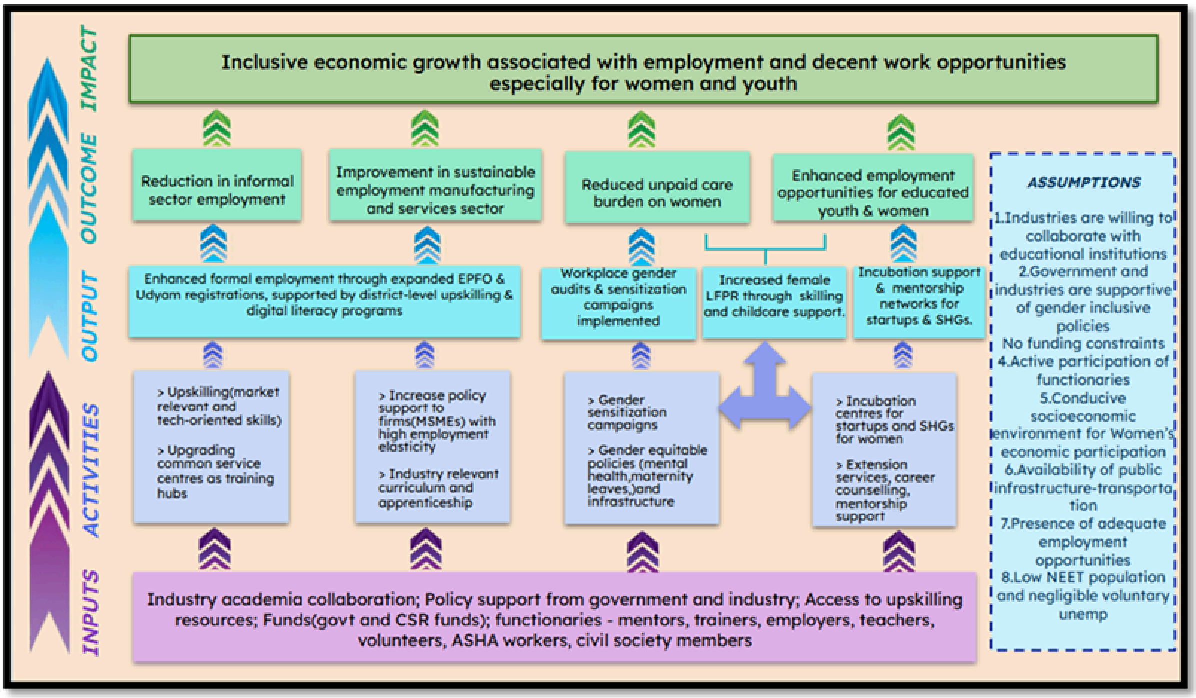
3. **Inclusivity**
   Inclusive labor markets ensure equitable participation across gender and age groups. The *Labour Force Participation Rate (LFPR)* and *Worker Population Ratio (WPR)*, disaggregated by gender and youth, assess participation levels, while *Women's* and *Youth Employment Shares* indicate the extent of empowerment and social inclusion.

4. **Productivity**
   The *Labour Productivity Growth Rate* captures the efficiency of converting labor inputs into output, reflecting improvements in skills, technology, and organization. Rising productivity is essential for sustained income growth and macroeconomic stability.

5. **Socio Economic Outcomes**
   Employment ultimately influences broader development outcomes. The *Poverty Rate*, *Average*

Income, *Access to Education*, and *Life Expectancy* together measure the long term social and economic impact of labor market performance.

INDICATIVE THEORY OF CHANGE



DATA AND VARIABLES USED

International Data

The data for international-level insights were obtained from the World Bank's World Development Indicators (WDI) for the year 2022, encompassing 217 countries worldwide. For this study, the primary focus has been placed on developing economies, particularly Southeast Asian countries and G20 member states, with comparisons drawn against developed economies to highlight structural contrasts.

Key Indicators Analysed

- GDP per capita, PPP (constant 2021 international $)
- Agriculture, Forestry and Fishing Value Added (% of GDP)
- Industry (including Construction) Value Added (% of GDP)
- Services Value Added (% of GDP)
- Employment in agriculture (% of total employment) (modeled ILO estimate)
- Employment in industry (% of total employment) (modeled ILO estimate)
- Employment in Services (% of total employment) (modeled ILO estimate)
- Vulnerable Employment (% of total employment) (modeled ILO estimate)

The year 2022 was chosen as the reference period for all indicators due to its comprehensive data availability, post-COVID-19 economic stabilization, and relevance for current global development trends.

Data Treatment Framework

To ensure analytical rigor and comparability, the following data treatment steps were undertaken:

- **Missing Value Imputation:** Missing observations were imputed using a bivariate anchor-variable approach, with GDP per capita (PPP) serving as the anchor. This method helped preserve the internal consistency of the dataset while maintaining comparability across countries.
- **Transformations:** Log transformation was applied exclusively to the GDP per capita (PPP) variable to address skewness and ensure robust statistical interpretation.
- **Capping:** Outliers were identified and capped at the 5th and 95th percentiles for most indicators. For Agriculture Value Added, Employment in Agriculture, and Vulnerable Employment, capping was applied at the 10th and 90th percentiles to better reflect the distribution of values without any distortion.

This data treatment framework ensures robustness, reliability, and comparability of the dataset, enabling objective and empirically sound insights throughout the analysis.

National Data

The analysis draws upon two principal national-level datasets:

- **National Account Statistics (NAS)**: Provides comprehensive macroeconomic indicators, including sectoral share and growth rates of Gross Value Added (GVA) across Agriculture, Industry, and Services sectors for the period 2011–12 to 2023–24.

- **Periodic Labor Force Survey (PLFS) Annual Reports (2017–18 to 2023–24)**: Offers disaggregated data on labor force participation, workforce participation ratios, unemployment rates, employment status distribution, industry-wise employment (NIC 2008 classification), and salary quality indicators. Special attention was given to demographic sub-groups—Women and Youth (aged 15–29 years)—to assess inclusive growth and sectoral challenges.

Key Indicators Analysed

- Sectoral share and growth rate of GVA (2011–12 to 2023–24)
- Labor force participation rate (usual status, ps+ss)
- Workforce participation ratio (usual status, ps+ss)
- Unemployment rate (usual status, ps+ss)
- Worker distribution by status in employment (usual status, ps+ss)
- Employment by industry type (NIC 2008 classification)

- Salary quality attributes

Data Treatment Framework

- **Youth Data Gaps:** PLFS reports do not consistently report youth (15–29) employment breakdowns by type, industry, or salary. To address this, indicators such as 'employment type' and 'industry type' for youth were synthetically constructed using statistical techniques in R programming.
- **Missing Yearly Data:** Formatting inconsistencies across PLFS reports impeded direct year-on-year comparisons. Specifically for 2022–23, employment type data for youth was not available. Univariate Missing Value Imputation (using mean values from adjacent years) was employed to estimate missing figures.
- **Salary Quality Estimation:** While basic mathematical deductions were made, econometric and advanced estimation techniques present opportunities for greater precision—though outside the scope of the current study.

State Data

Analysis at the state level was based on:

- **PLFS Annual State-Level Data (2017–18 to 2023–24):** Includes state-wise unemployment rates (for youth, females, overall), sectoral employment breakdowns, employment type (self-employed, regular wage/salaried, casual labor), and salary quality indicators (job contracts, paid leaves, social security benefits).

State Indicators Analysed

- State-wise unemployment rates for youth, females, and total population
- Sectoral employment structure (agriculture, industry, services)
- Employment type distinctions
- Salary quality and job benefits

Data Treatment Framework

- **Consistent Issues:** State-level PLFS analysis encountered similar challenges as national data. Youth-specific employment type indicators for 2022–23 were unavailable, but since state-level comparisons focused on multi-year trends (2017–18 to 2023–24), isolated gaps had limited impact on overall analysis.
- **Youth Data Limitations:** Detailed employment information for youth (15–29) was not reported for several indicators, restricting youth-specific analysis at the state level.

METHODOLOGY

This white paper undertakes a multi-layered approach for a holistic assessment of India's growth trajectory and structural challenges at the global, national and regional level.

At the global level, a comparative multivariate analysis of India's economic growth trajectory is undertaken, employing both trend analysis and scatter plot visualization to elucidate patterns across countries. Sectoral transformation within India is further benchmarked against select regional peers—including Bangladesh, Vietnam, Sri Lanka, and South Africa—to facilitate meaningful cross-country comparison and highlight distinctive features of India's structural transformation.

At the national level, key macroeconomic indicators—including GDP growth, sectoral composition, and labour market statistics—are analysed to examine the interplay between economic growth and employment generation. Trend analysis and descriptive statistics are employed to trace structural shifts, cyclical patterns, and deviations over time.

At the regional level, Indian states are compared across multiple parameters of jobless growth to capture inter-state disparities. A state-level Quality of Growth (QoG) Index is developed to evaluate the nature and inclusiveness of growth over the past seven years.

QoG Index

It is a composite measure designed to capture the multidimensional nature of growth across Indian states. It incorporates key parameters such as the unemployment rate, sectoral employment distribution, employment type, salary quality, and Gross State Domestic Product (GSDP). These indicators were carefully selected to present a holistic picture of growth performance and inclusiveness.

The analysis covers the period between 2017 and 2022, examining changes across three demographic groups: the overall population, women, and youth. The inclusion of gender- and youth-specific indicators enables a focused assessment of demographic patterns within state-level growth.

To construct the QoG Index, all indicators were first categorized as positive or negative based on their developmental implications. For instance, higher unemployment rates, greater employment in agriculture, and a larger share of informal employment without social protection were classified as negative, whereas higher employment in the services sector and higher GSDP were considered positive. This classification facilitated the quantification and standardization of indicators on a comparable scale.

Using the Min-Max normalization method, all variables were rescaled between 0 and 1 to ensure consistency across states. The Unweighted QoG Index was then computed as the arithmetic mean of these normalized values. To validate the robustness of the index, a sensitivity analysis was conducted by randomly assigning weights to the indicators and comparing the results with the unweighted index. This approach ensured the comprehensiveness and reliability of the QoG Index within the available data framework.

KEY INSIGHTS

Insights at the international level

To make sense of India's economy, study of its similarities as well as uniqueness in the global scenario is non - negotiable.

Global trends in the contribution of agriculture and services to GDP

Globally, the structural composition of GDP aligns with the established development theory, that is, as economies advance, the relative contribution of agriculture declines while that of services expands. Figure 1 highlights an inverse relationship between agricultural value added (Y-axis) and GDP per capita (X-axis), underscoring the declining role of agriculture with rising income levels. Conversely, Figure 2 depicts a positive correlation between the contribution of services (Y-axis) and GDP per capita (X-axis), with an upward-sloping trend line reflecting the growing dominance of the services sector as GDP surges. Taken together, these patterns reaffirm that for most growing economies, structural transformation is characterized by a steady transition from agriculture to services.

Figure 3: Agriculture (% of GDP) 2022

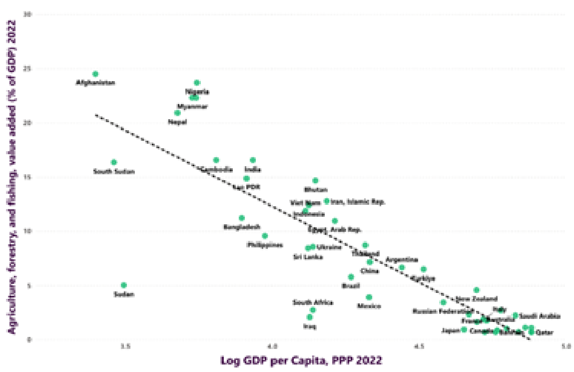
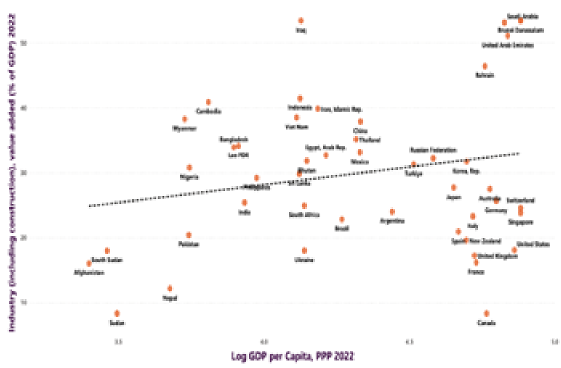


Figure 4: Industry (% of GDP) 2022



However, as shown in Figure 3, India stands noticeably above the global trendline, indicating that its agriculture sector accounts for a larger share of GDP than expected for its level of per capita income. While countries with comparable or higher incomes—such as Bangladesh, Vietnam, and the Philippines—typically record agricultural shares in the 11–12% range, India's remains at 16.5%.

In contrast, India's industrial sector (including construction) lies below the trendline as in Figure 4, contributing 25.4% of GDP. This is lower than in economies like Vietnam, Bangladesh, and Indonesia, which have achieved higher industrial shares in the 33-39% range at similar income levels.

Meanwhile, India's services sector, at 49.5% of GDP, aligns closely with the trendline for its income level and mirrors the shares observed in Bangladesh at 51%. Regional peers such as Vietnam and the Philippines display similar services proportions when adjusted for GDP per capita, whereas higher-income economies like Singapore and Japan show substantially larger services shares—an outcome of more advanced structural transformation.

In essence, India's sectoral composition remains more agriculture-heavy and less industry-driven, even as its services sector keeps pace with regional counterparts.

Comparing Sectoral Share of GDP Per Capita 2022

Country	GDP per capita	Agriculture (% of GDP) 2022	Industry (% of GDP) 2022	Services, value added (% of GDP) 2022
Brazil	**18554.0**	**5.8**	**22.8**	**58.1**
China	**21499.4**	**7.1**	**37.9**	**55.0**
Bangladesh	**7888.2**	**11.2**	**33.9**	**51.0**
India	**8594.4**	**16.5**	**25.4**	**49.7**
Turkiye	**32748.3**	**6.5**	**31.3**	**51.7**
Viet Nam	**12979.8**	**11.9**	**38.5**	**41.1**

Global trends in the employment share

Employment in agriculture: The graph shows relationship between GDP per capita (X-axis, $\log_{10}$ scale) and Employment in agriculture as a percentage of total employment (Y-axis) for 217 countries in 2022. This graph supports the general economic theory that as countries develop and GDP per capita increases, employment shifts away from agriculture towards industry and services. This indicates that countries with higher GDP per capita tend to have a lower percentage of employment in agriculture.

India's position: India's employment in agriculture is 42.86% for a GDP per capita of $8,594 ($\log_{10}$ value of 3.93). For this income level, the trendline predicts agriculture employment of 29.33%, meaning India's actual value is +13.53 percentage points above the trendline. Compared to similar economies with a peer average of 23.45%, India's agriculture employment is +19.41 percentage points higher. Even Bangladesh (GDP $7,888) has lower agricultural employment at 35.66%. This indicates that India has significantly more workers engaged in agriculture than expected for its income level, suggesting substantial scope to transition workers to more productive sectors.

Employment in industry: The relationship between GDP per capita (X-axis, $\log_{10}$ scale) and Employment in industry as a percentage of total employment (Y-axis) for 217 countries in 2022. The graph shows a weak positive correlation. This suggests that as GDP per capita increases, there is a modest tendency for industrial employment to rise, though with significant variation across countries.

India's Position: India's employment in industry is 26.12% for a GDP per capita of $8,594 ($\log_{10}$ value of 3.93). For this income level, the trendline predicts industrial employment of 18.18%, meaning India's actual value is +7.94 percentage points above the trendline. Compared to similar economies with a peer average of 21.77%, India stands +4.35 percentage points higher. This indicates that India has a stronger industrial employment base than expected for its income- level, suggesting ongoing industrialization with manufacturing and construction sectors absorbing a larger share of the workforce than most comparable countries.

Employment in services: The relationship between GDP per capita (X-axis, $\log_{10}$ scale) and Employment in services as a percentage of total employment (Y-axis) for 217 countries in 2022. The graph aligns with the general economic pattern where developed countries have a higher share of employment in services. The trendline shows a positive relationship, indicating that as GDP per capita increases, the share of services employment also tends to rise.

India's Position: India's employment in services is 31.02% for a GDP per capita of $8,594 ($\log_{10}$ value of 3.93). At this income level, countries typically have much higher services employment, often exceeding 50%, indicating that India's service sector employment is underdeveloped relative to its income level. Compared to similar economies with a peer average of 54.78%, India lags by -23.76 percentage points. Even Bangladesh ($7,888) has 43.79% services employment, while Jordan ($9,267) has achieved 78.37%. This indicates that despite India's growing service sector GDP contribution, the sector is not absorbing a proportionate share of the workforce, suggesting substantial scope for India to create more service sector jobs and transition workers from agriculture to services, like the employment patterns seen in peer nation.

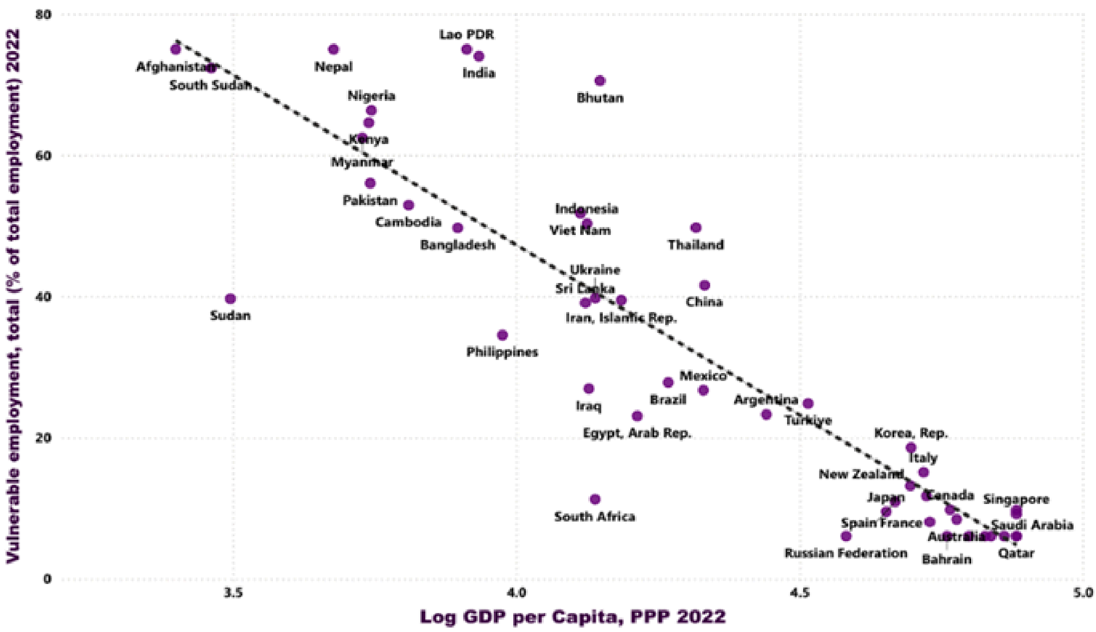
Comparing Sectoral Share of Employment 2022

Country	Employment in agriculture (% of total employment) 2022	Employment in industry (% of total employment) 2022	Employment in services (% of total employment) 2022
Brazil	8.7	20.5	70.8
China	22.8	31.7	45.6
Bangladesh	35.7	20.5	43.8
India	42.9	26.1	31.0
Türkiye	15.7	27.7	56.6
Viet Nam	33.5	30.7	35.9

The quality of employment – vulnerable employment

India’s vulnerable employment rate remains among the highest globally (~73% as per ILO/World Bank 2022, well above the LMIC average of 53%). The best-fit line on the scatterplot shows that as GDP rises across countries, vulnerable employment falls steeply—but India lies well above this line, signalling its growth is not translating into formal, secure jobs for most of its population.

Figure 9: Vulnerable employment vs GDP per capita, 2022



Despite reaching middle-income status, India’s vulnerable employment rate is much higher than what would be predicted for its GDP per capita—even higher than regional and global peers with similar or lower income. For example, Vietnam, Indonesia, Bangladesh and Sri Lanka all have 20–35 percentage points less vulnerable employment at similar income levels. This signals that India's growth has not been accompanied by sufficient improvement in job quality or formalization, and it stands out as a negative outlier globally and in South Asia.

India’s agricultural employment share is 43%—the highest outside sub-Saharan Africa—and its vulnerable employment mirrors this (near 74%). This is not merely a developmental lag; it’s a demonstration of “**distress-driven absorption**”: as manufacturing fails to create enough jobs, women and youth are absorbed into unpaid family farm work, which is systematically non-contractual and unprotected. India as a Global Outlier: High Vulnerability Despite Structural Change in GDP. It reflects a paradox: services now comprise nearly 50% of Indian GDP, yet vulnerable employment hasn’t fallen as in peer economies.

Insights at the National Level

Sectoral breakdown of GVA

Figure 10

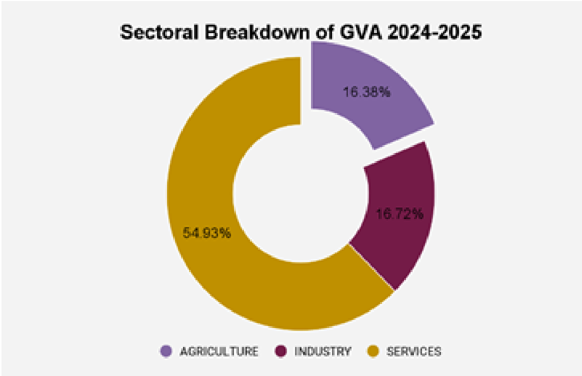
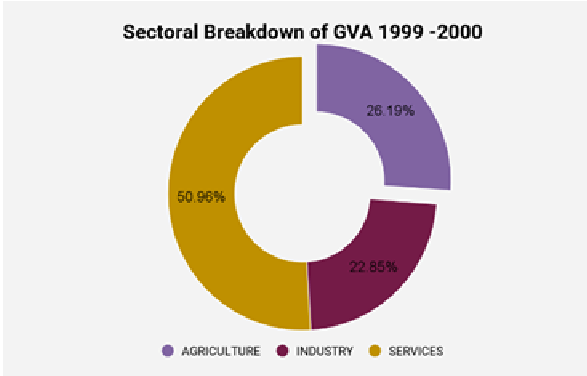


Figure 11



The sectoral GVA composition shows services increasing from 50.96% (1999-2000) to 54.93% (2024-2025), cementing its position as the dominant economic driver. However, services employ only approximately 30% of India's workforce despite contributing over half of economic output. The services sector is employing a relatively small, educated workforce. Meanwhile, 46.1% of India's workforce remains locked in agriculture, which contributes merely 16.38% to GVA (2024-25). This 3:1 employment-to-GDP ratio in agriculture versus the inverse ratio in services exemplifies jobless growth—where GDP expansion bypasses much of the labor force. The "Missing Middle" of Structural Transformation - Despite 25 years of economic reforms, manufacturing's GVA share has shown only marginal movement from 22.85% (1999-2000) to 18.72% (2024-2025), and alarmingly, manufacturing employment has stagnated at 11-12% of the total workforce. The graphs document what development economists call "premature deindustrialization"—where countries transition to service-dominated economies before achieving the manufacturing-led employment growth that historically lifted populations out of poverty. The sectoral trend chart (2011-2024) reveals services maintaining a dominant 52-55% GVA share with consistent 6-8% growth rates, yet this growth is structurally inaccessible to many job seekers.

Sectoral employment share

While youth non-agricultural employment stands at only 31.3% (industry) + 29.5% (services) = 60.8% in 2023-24, and female non-agricultural employment at just 20.12% (industry) + 15.54% (services) = 35.66%, this represents a massive underrepresentation relative to sectoral GDP contributions (where industry + services account for 84% of GVA). This means youth and especially women are locked out of the sectors generating economic value. Sectoral employment graphs show agriculture absorbing 46.07% of total employment (2023-24), yet women constitute 64.4% of agricultural workers, a proportion that has increased from 57% (2017-18) to 64.4% (2023-24) : it represents "feminization of agriculture". The Labor Force Participation Rate (LFPR) graphs confirm this: female LFPR rose from 53% to 64.9% (2017-2024), but this increase is concentrated in rural areas (+23 percentage points rural vs +5 points urban).

Figure 12: Employment trends in agriculture

Source: PLFS

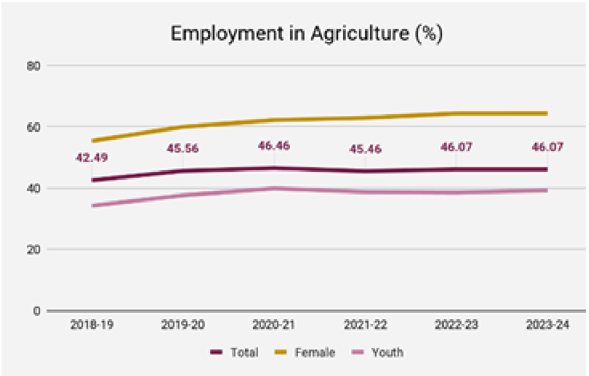
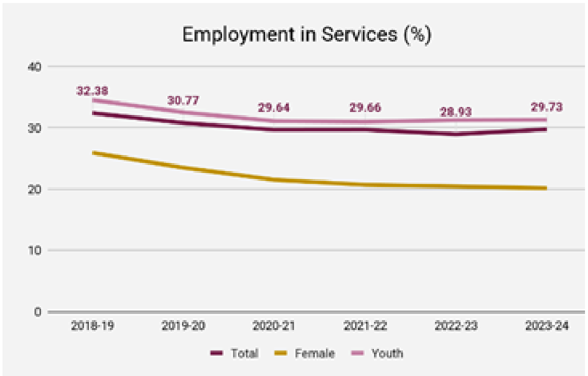


Figure 13: Employment trends in services

Source:
PLFS



Shows services employment declining from 32.38% (2018-19) to 29.73% (2023-24)—a 2.65 percentage point drop even as services GDP share increased over same time.

The LFPR and WPR graphs show impressive increases: total LFPR from 53% to 64.9%, female LFPR from 23.3% to 41.7% (2017-24), appearing as evidence of inclusive growth. However, unemployment rates simultaneously declined from 6.5% to 3.2%, suggesting near-universal employment. This is statistically inconsistent with stagnant sectoral employment shares.

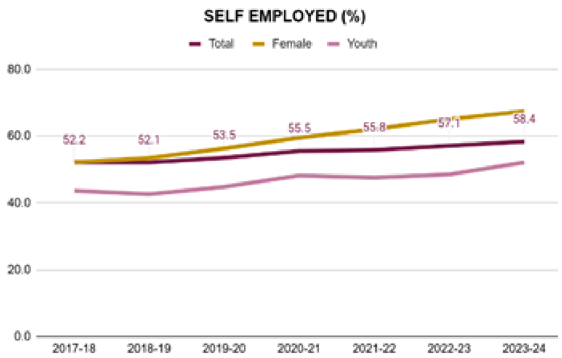
Employment type

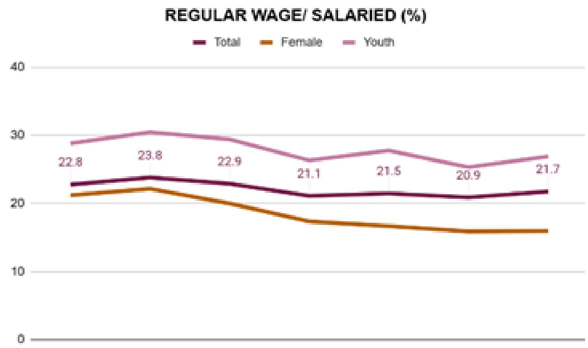
Figure 14: Self-employed trend 2017-18 to 2023-24

Figure 15: Regular wage trend 2017-18 to 2023-24

Source: PLFS

Source: PLFS





The charts show an upward trend in self-employment for all groups, especially among youth (rising to 59.4%) and women (approaching 55%) by 2023-24. However, this increase is overwhelmingly driven by a surge within household enterprises. Despite a growing economy, the proportion of the workforce engaged in regular wage/salaried jobs remains weak and stagnant (total: about 21%, female: near 15%, youth: about 17%). This is nearly unchanged since 2017-18 or has even dipped in some years. This highlights a persistent failure to create quality, secure employment, especially in manufacturing and services. Decline in casual employment reflects shift to unpaid family work rather than better job quality. Gendered nature of vulnerability — Women's "Self-Employment" is overwhelmingly unpaid, intensifying time poverty. Although overall employment levels and participation rates may rise, India's employment structure is tilting ever more toward informal, unpaid, or precarious roles, especially for young people and women. The rise of "self-employment" is a symptom of labor market distress and a shortage of decent, formal jobs.

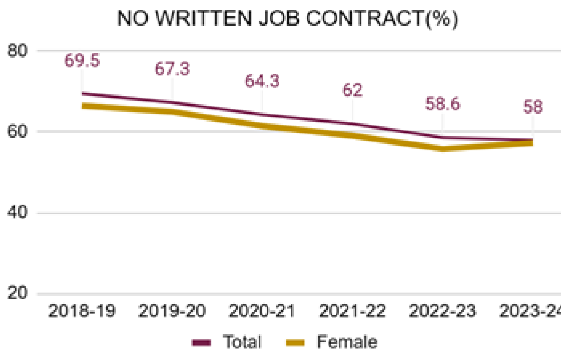
Employment quality

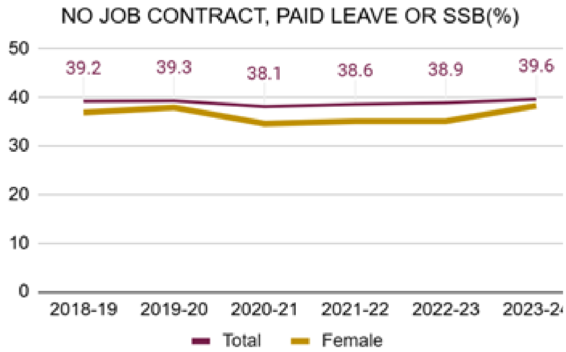
Figure 17: Employment trends for no written job contract

Figure 18: Employment trends for no job security

Source: PLFS

Source: PLFS





About 47.3% of all non-agricultural workers, including 45.9% of women and 48% of youth, were not eligible for paid leave in 2023-24. This signals widespread non-compliance with minimum labor standards. The data show 58% of non-agriculture workers had no written job contract in 2023-24, with women (57.3%) and youth (58%) similarly affected. Despite a declining trend (down from 69.5% in 2017-18), informality remains extremely high, exposing workers to arbitrary wage practices, lack of job security, and easy dismissal. The percentage with access to any social security benefit lingers below 54% for all groups in 2023-24 —A “Protection Gap” for Young and Female Workers. The “all three above” metric reveals 39% of non-agricultural workers (and 38.3% of women and youth) face triple exclusion—no contract, no paid leave, no social security—showing very slow improvement over time.

Insights at the State Level

India’s economic growth trajectory is increasingly determined by variations at the state level, shaped by both constitutional autonomy and local governance dynamics. States wield significant power over agriculture, industry, and labour, and their fiscal scale now surpasses the Union government, demanding granular monitoring of economic and employment performance. Policy debates must extend beyond headline growth, examining whether states convert expansion into broad-based job creation and improved work quality.

Quality of Growth Index

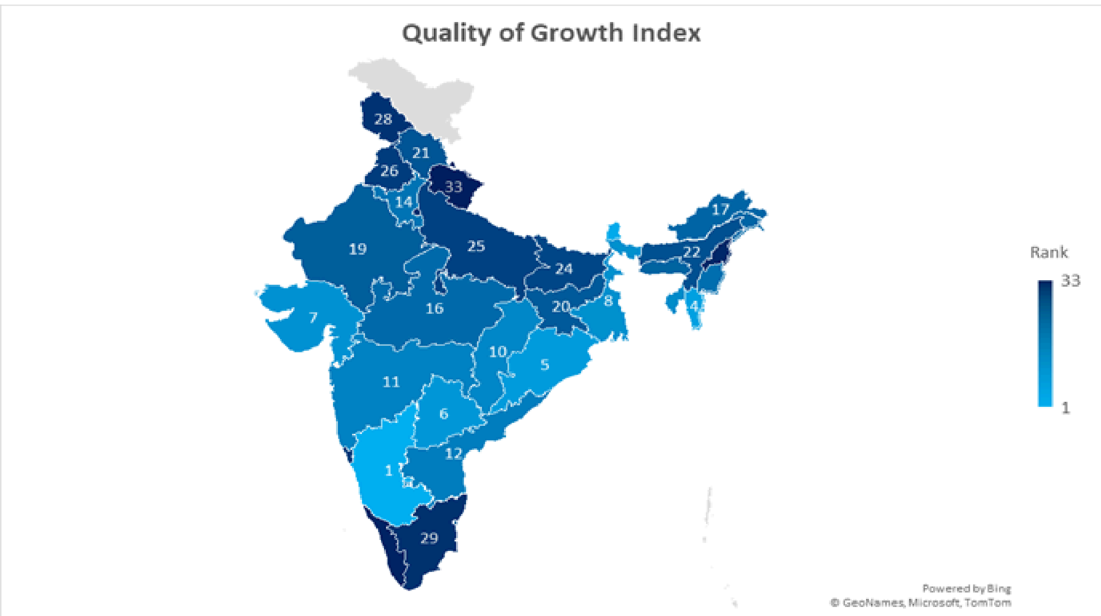
The Quality of Growth Index offers a composite lens to track economic transformation at the subnational level, assessing 18 multidimensional indicators grouped under five broad dimensions: unemployment reduction, employment type transitions, sectoral shifts, job quality and security, and five-year per capita income growth. The rigor in its construction—normalizing and integrating diverse measures ensures comparability and reveals qualitative gaps invisible in GDP figures.

QoG Index Score and Rankings

The rankings reveal that states such as Karnataka and Telangana, which have demonstrated strong digital governance, innovation ecosystems and investment-friendly policies, performed exceptionally well. These states achieved notably strong composite scores, driven particularly by the performance in sectoral employment shifts and unemployment reduction indicating substantial workforce transition from agriculture to higher-productivity sectors coupled with a perfect score in five-year PCI growth, reflecting both employment quality improvements and robust income expansion. In contrast, Bihar, Uttar Pradesh and Nagaland, which continue to face structural challenges such as low industrial diversification, weak infrastructure, and limited formal employment, ranked significantly lower.

Figure 19: QoG Index Rankings for States/UTs

Source: PLFS



Interestingly, Odisha emerged as a strong performer, securing 5th rank, signalling significant governance improvements and sustained efforts in industrial and social infrastructure expansion despite historically being overlooked in mainstream development discourse. Meanwhile, states like Kerala, long praised for their human development and social indicators, performed poorly in this ranking. This may be attributed to slower structural transformation, high reliance on the service sector and limited creation of new formal jobs.

Overall, these rankings underscore the need for policymakers and analysts to reassess traditional measures of development to have a clear picture of the economic transformation happening at the state level.

Comparing Per Capita Income growth rate and QoG Index of the states/UTs

The comparative ranking between Quality of Growth and Per Capita Income reveals important structural and governance differences across Indian states.

1. High Performance on Both Dimension

States like Karnataka, Telangana, Mizoram and Gujarat exhibit both high-quality growth and high per capita income. These states have combined rapid industrialization, digital innovation and service-sector dynamism with governance reforms. They reflect a balanced growth model, where economic expansion is accompanied by improvements in institutional and social outcomes.

2. Strong Growth Quality Despite Moderate Income

States such as Haryana, Odisha and Tripura show strong quality of growth even though their per capita income levels are relatively modest. This points to inclusive and sustainable growth patterns, possibly due to effective social programs, governance reforms and investment in human capital. Odisha’s performance is particularly notable, as it is rarely highlighted in mainstream growth discussions yet has shown consistent institutional improvement. Haryana's better performance in quality of growth can be attributed to its leading position in the employment type indicator, accompanied by favourable outcomes in unemployment, social security benefits (SSB) and job quality indicators.

3. Economically Prosperous but Weak in Growth Quality

Delhi, Goa and Kerala demonstrate high per capita income but low quality of growth rankings. These states may have mature economies with strong service sectors but show signs of stagnation in new job creation or industrial diversification. The lower quality scores reflect limited structural transformation, over-dependence on tertiary sectors and possibly high inequality or informality. A key factor behind this is poor performance in sectoral employment shifts, employment type improvements, and social security expansion. This trend highlights that high GDP growth does not necessarily translate into qualitative improvements in employment conditions.

4. Low Performance on Both Dimensions

States such as Bihar, Uttar Pradesh, Nagaland and Uttarakhand lag on both parameters. These regions continue to face low productivity, weak governance capacity, and inadequate industrial infrastructure, leading to both low income and poor-quality growth.

From the above observations, it can be noted that the GDP growth by itself does not ensure equitable and sustainable economic transformation. The QoG index analysis emphasises the significance of taking employment quality, job security and job creation into account when assessing India's economic success.

Figure 20: GDP Per Capita Growth Rate Rankings of States/ UTs

Source:
PLFS

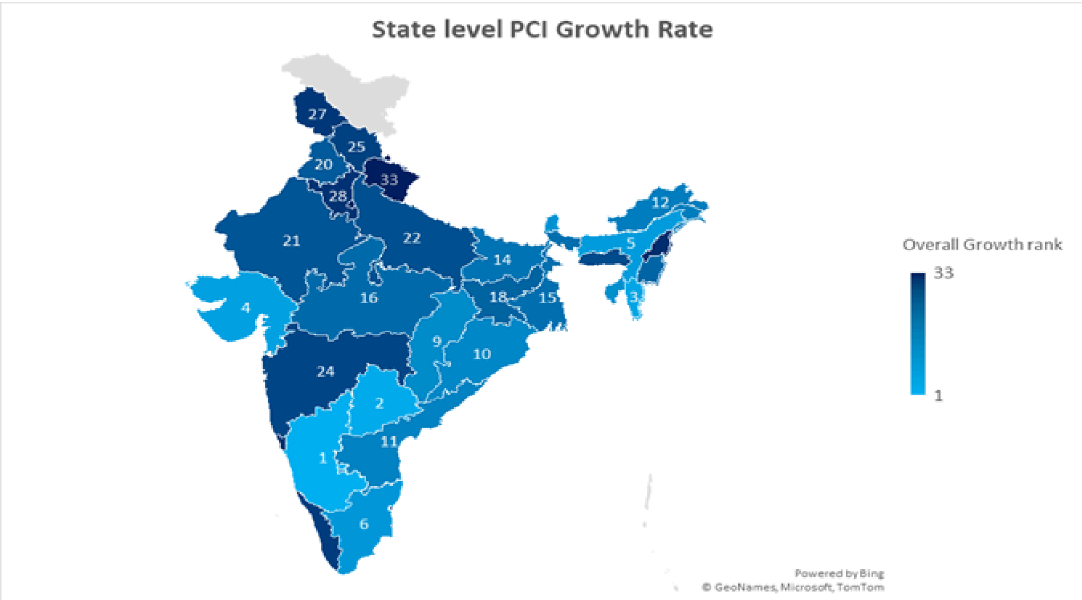
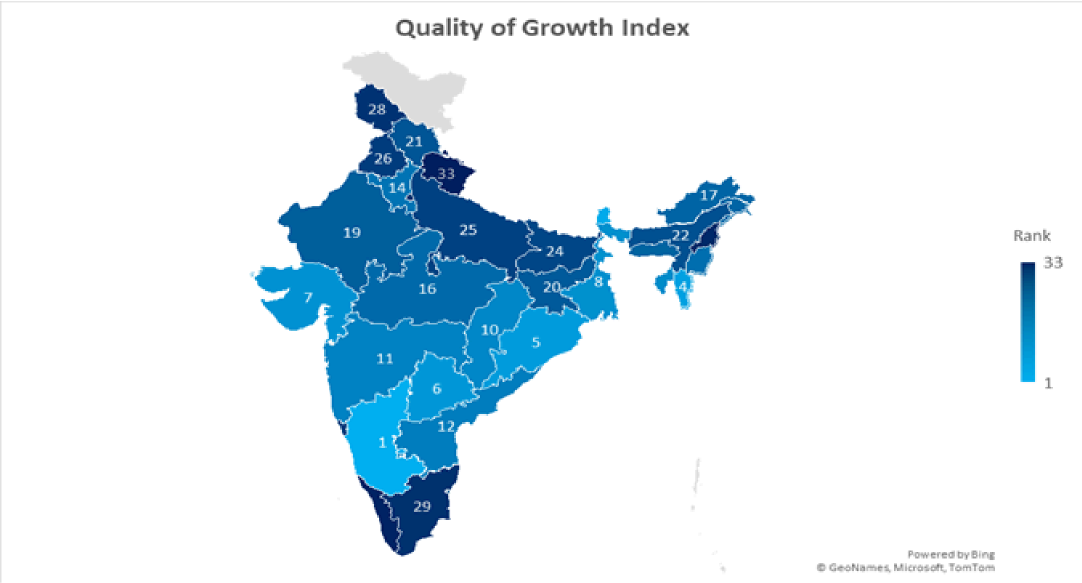


Figure 21: QoG Ranking of States/UTs

Source: PLFS



POLICY SIMULATIONS

Employment Elasticity

Year	Employment Elasticity	Agriculture Elasticity	Industry Elasticity	Services Elasticity
2017	–	–	–	–
2018	0.44	-0.55	0.66	0.87
2019	2.42	2.16	-1.27	-1.03
2020	-0.9	-0.51	-0.33	0.47
2021	3.51	-0.14	0.52	0.09
2022	-2.85	0.28	0.16	-0.22
2023	0.79	0.1	-0.25	0.4

How each column was calculated:

1. **Absolute Employment Number**
   - Formula:

$$\text{Absolute Employment} = \text{Population} \times \frac{\text{WPR}}{100}$$

   - Example (2018):
     $1,374,659,064 \times (35.3/100) = 48,525,646.96$
2. **Employment Growth Rate** (percentage change over previous year)
   - Formula:

$$\text{Employment Growth Rate} = \frac{\text{Employment}_{\text{year}} - \text{Employment}_{\text{year before}}}{\text{Employment}_{\text{year before}}} \times 100$$

   - Example (2018):

$$\frac{48,525,647 - 47,180,118}{47,180,118} \times 100 = 2.85\%$$

3. **Employment Elasticity**
   - Formula:

$$\text{Employment Elasticity} = \frac{\text{Employment Growth Rate}}{\text{GDP Growth Rate}}$$

   - Example (2018):

$$\frac{2.85}{6.45} = 0.4421$$

With elasticity consistently below 1, every 1% of GDP growth is generating less than 1% new jobs. Raising elasticity through apprenticeships, employer-led skilling, and active placement programs could realistically shift industry and services elasticity into the 0.7–1.0 zone—critical for matching job creation to India's growing labor force.

Is India's Growth Jobless?

Current employment elasticity trends confirm that India's economic growth has not been consistently job-rich:

- GDP and sectoral output have grown, but employment has not kept pace, especially in manufacturing and agriculture.
- Extended periods of negative elasticity mean job losses occurred even as output increased.

Comparative Benchmark

- Peer economies (China, Vietnam, Indonesia, Sri Lanka) sustained national elasticities above 0.7–1.0 during phases of inclusive growth.
- India's national average lags, reinforcing the need for focus on job creation, not just output expansion.

<u>**Reasons for taking elasticity as benchmarks:**</u>

1. International Benchmarks: Countries with "growth-with-jobs" records

- East Asian economies (China, Korea, Taiwan) during their demographic dividend had aggregate national elasticity around 0.7 -- 1.0 during their rapid, inclusive growth periods (World Bank, ILO studies).
- OECD and Latin American examples: Most countries with elasticities above 0.7 saw noticeable year-on-year declines in unemployment, especially when sectoral elasticities were above 0.5 in industry and 0.8 in services.

2. Indian Historical and Comparative Trends

- NITI Aayog (2017, 2022 reports) and India Employment Reports recommend raising overall elasticity to 0.7–1.0 for actual livelihoods improvement.
- Indian economic literature (e.g., Suresh Tendulkar, Santosh Mehrotra) uses 0.7–1.0 as a normative benchmark to support inclusive growth with the expanding labor force.

3. Logical and Quantitative Requirement

- Elasticity of ~1: Jobs rise 1% for every 1% GDP growth = “jobful” growth.
- Elasticity below 0.5: Even high GDP growth rapidly fails to reach demographic and employment targets (leads to “jobless growth”).

5. Sectoral Global Patterns

- Agriculture generally creates fewer jobs per % GDP growth (due to mechanization, surplus labor)—thus, 0.2–0.4 is a realistic positive target.
- In industry and services, elasticities above 0.7 have empirically supported rising employment in almost all “East Asian Miracle” and successful growth-with-jobs cases.

Achieving Growth Through Employment Elasticity (2024–2047)

Future Trend Projections:

- Based on historical trends and policy optimism, India's national employment elasticity is projected to rise gradually from its current value of 0.3 in 2024 to 0.7 by 2047, with a steady GDP growth rate of 6.5%.
- This approach is designed to be both feasible and ambitious—accounting for structural change, gradual sectoral reform, and improved cross-sector job quality.

Year-on-Year Calculations:

- Employment elasticity improves incrementally each year as reforms deepen and inclusion rises.
- GDP growth rate is fixed at 6.5%, in line with policy consensus and historical performance.
- Employment growth rate is directly calculated as:

$$\textbf{Employment Growth Rate (\% per year) = Employment Elasticity} \times \textbf{6.5}$$

- For example:
  2024: 0.3 × 6.5 = 1.95%
  2030: 0.42 × 6.5 = 2.74%
  2047: 0.7 × 6.5 = 4.55%

Year	Projected Employment Elasticity	Assumed GDP Growth Rate (%)	Implied Employment Growth Rate (%)
2024	0.30	6.5	1.95
2028	0.37	6.5	2.40
2036	0.53	6.5	3.46
2040	0.60	6.5	3.90
2047	0.70	6.5	4.55

Year | Projected Employment Elasticity | Assumed GDP Growth Rate (%) | Implied Employment Growth Rate (%)

Policy Simulations and Recommendations by Sector

Agriculture

- Action: Scale up agro-processing clusters, invest in high-value crops, and build rural non-farm job engines via MSMEs and skill missions.
- Projection: Sector elasticity rises from ~0.2 to 0.4 by 2047. Rural employment transitions toward higher productivity and income, reducing vulnerable jobs.
- Recommendation: Target annual employment growth of 1–2% in agriculture, matching GDP shifts and labor exits with new livelihoods.

Industry

- Action: MSME digitalization, fiscal incentives for job-rich sectors (garments, food processing, electronics), mandatory apprenticeships, and cluster formalization.
- Projection: Elasticity increases from 0.3 to 0.7; industry provides 4–5% annual job growth by 2047.
- Recommendation: Support 8–12 million quality jobs each year; raise industry share in formal employment.

Services

- Action: Massive skilling, digital job platforms, healthcare/care-sector expansion, and gender/youth inclusion in IT and urban retail.
- Projection: Elasticity rises from 0.4 to 0.8+, delivering 5%+ annual job growth.
- Recommendation: Ensure majority share of new jobs is formal, accessible to marginalized populations.

National (Aggregate)

- Action: Quarterly labor market dashboards, E-Shram-linked protections, cross-sector migration management, benefit portability, and digital contract enforcement.
- Projection: National elasticity rises to 0.7, producing employment intensive growth rates that reliably outpace labor force expansion.
- Recommendation: Anchor policy in real-time elasticity data; adapt strategy yearly to secure inclusion and job creation.

LIMITATIONS

1. Methodological Constraints

The construction of the Quality of Growth Index (QGI) involves inherent methodological subjectivity. The predetermined weights (0.15, 0.25, 0.20, 0.10), though consistent with established practice, rely on normative judgment; alternative weighting or aggregation methods could yield different rankings. Linear aggregation assumes perfect substitutability across dimensions, overlooking non-compensatory relationships among growth, inclusion, and employment quality. Normalization procedures such as min–max or z-score are sensitive to outliers, potentially distorting relative performance. Furthermore, missing value imputations, while necessary for completeness, introduce uncertainty and possible bias, especially where interstate data availability is uneven.

2. Data Quality and Availability

The robustness of findings is limited by persistent data deficiencies. The diversity and fluidity of India's informal sector mean that many employment types such as migrant, seasonal, and gig or platform-based work evolve faster than official surveys can capture, producing a snapshot rather than a continuous understanding of labor dynamics. With over 80 percent of India's workforce employed informally, this underrepresentation significantly affects measurement accuracy. Women's unpaid and subsistence work is similarly excluded, leading to underestimation of female labor participation. Youth transitions between education, unemployment, and employment are poorly reflected in cross-sectional surveys. Smaller states and Union Territories often face limited or inconsistent data, while temporal lags in key sources such as the PLFS reduce the timeliness of analysis.

3. Conceptual and Theoretical Issues

Condensing multidimensional employment and growth outcomes into a single index risks oversimplification. Strong performance in one domain may conceal weaknesses in another, reducing interpretive clarity. Indicator and weight selection remain open to normative or political contention. Moreover, the QGI establishes correlation rather than causation between governance, growth, and employment performance, limiting its explanatory depth.

4. Scope and Coverage

The index does not capture intrastate disparities such as rural and urban, caste, and gender-based inequalities, and only partially reflects job quality dimensions like wages, security, and working conditions. Sectoral aggregation further conceals variation across industries, while limited gender and youth disaggregated data constrain inclusivity assessment.

5. Benchmarking and Comparability

Divergence between QGI results and other composite indices such as NITI Aayog's SDG Index stems from differing conceptual frameworks and indicator priorities, which may cause interpretive ambiguity for policymakers. Additionally, the absence of international benchmarks restricts assessment of state performance against global standards.

6. Policy Relevance and Practical Utility

While the QGI provides a holistic overview, it offers limited diagnostic precision for targeted interventions. As a static measure, it does not account for temporal shifts or improvement trajectories. Aggregate state level rankings obscure heterogeneity across districts, sectors, and demographic groups, reducing their direct policy applicability.

7. Structural Transformation and Implementation Feasibility

The index does not directly measure the pace or elasticity of structural change, such as labor movement from low to high productivity sectors, or productivity growth within informal work. Aggregating all services masks disparities between high value sectors like IT and finance and low productivity segments such as trade and transport. Moreover, the model's specification and weighting scheme lack sensitivity testing, limiting robustness and external validity. Translating findings into policy may be constrained by limited fiscal and institutional capacities, weak governance, and administrative inertia. Effective monitoring further depends on robust and disaggregated data systems, which remain inadequate in several states.

RECOMMENDATIONS

Based on the policy simulations and empirical findings, the following recommendations outline a pathway for creating inclusive, resilient, and job-rich growth in India by 2047. The focus is on improving the quality of employment through structural transformation, gender inclusion, social protection, and institutional coordination.

1. Agriculture: Enhancing Productivity and Rural Diversification

Agriculture continues to employ nearly half of India's workforce but contributes only a small share of national output due to low productivity. Simulations show that increasing investments in agro processing, food storage, and non-farm rural enterprises can raise employment elasticity from 0.2 to 0.4 by 2047.

Recommendations:

- Expand programs such as One District One Product and Farmer Producer Organizations to strengthen value chains.
- Integrate Krishi Vigyan Kendras with skill missions for training in processing and logistics.

- Link rural credit schemes with micro and small enterprise promotion to support non-farm livelihoods.

Expected Outcome: Balanced labor transitions from agriculture to higher productivity rural work and a one to two percent annual rise in rural employment.

2. Industry: Promoting Labor Intensive Manufacturing

Industrial employment can expand significantly through formalization and targeted incentives. Simulations suggest that focusing on labor intensive sectors could increase employment elasticity from 0.3 to 0.7 and generate up to five percent job growth each year.

Recommendations:

- Prioritize sectors such as textiles, food processing, and electronics through Make in India and Production Linked Incentives.
- Mandate apprenticeship programs and incentivize enterprises that hire women and young workers.
- Digitize small and medium clusters for easier compliance, access to credit, and e-market participation.

Expected Outcome: Creation of 8 to 12 million quality industrial jobs each year and wider formal employment coverage.

3. Services: Driving Inclusive Growth

The service sector shows potential for high productivity and inclusivity, with elasticity rising from 0.4 to above 0.8.

Recommendations:

- Invest in vocational and digital skills for healthcare, tourism, retail, and information technology.
- Expand the care and urban service economy to increase women's participation.
- Strengthen digital platforms for gig and freelance workers and integrate them into the E Shram database.

Expected Outcome: Higher female labor participation and more formalized service employment.

4. National Care Economy Accelerator

Drawing from international examples, India should establish a National Care Economy Accelerator with 10,000 community-based care clusters in districts where female self-employment is high.

Recommendations:

- Train and employ local women as caregivers and childcare workers at fair wages.
- Integrate care clusters with health and social protection missions.

Expected Outcome: Reduction in unpaid female labor and improvement in gender equality.

5. India SkillPass and Portable Benefits

A unified India SkillPass should record credentials, connect training to job placement, and automatically enrol workers in pension and insurance schemes.

Recommendations:

- Integrate Skill India, E Shram, and Aadhaar databases to create a single interoperable platform.

Expected Outcome: Broader social protection and improved labor mobility across states and sectors.

6. Universal Digital Social Security

A universal social protection system should integrate schemes such as Ayushman Bharat, PM Shram Yogi Mandhan, and Employee State Insurance into one digital framework.

Expected Outcome: Seamless and portable coverage for informal and vulnerable workers, particularly women and youth.

7. Institutional Coordination and Data Governance

A National Employment and Social Protection Council should be created to harmonize skilling, employment, and social security policies. Quarterly labor dashboards combining PLFS, CMIE, and E Shram data should guide policy revisions.

Expected Outcome: Evidence based and adaptive employment governance.

CONCLUSION

India's economic growth over the past decades has been strong, yet it has not fully translated into broad-based employment, particularly for women and youth. While the overall unemployment rate declined to 3.2 percent in 2022–23 according to the Periodic Labour Force Survey, youth unemployment remains high at 10 percent, and female labor force participation, though improving, is still around 37 percent. This indicates that India's growth is partially jobless: the economy is expanding, but the increase in output has not generated sufficient quality employment, especially for vulnerable and underrepresented groups. The disconnect reflects deep structural challenges including widespread informality, skill mismatches, and unequal access to opportunities across regions and sectors.

Bridging this gap requires a focus on employment-intensive and inclusive growth. Expanding agro-processing and rural non-farm enterprises, promoting labour-intensive manufacturing, and scaling up service sector skills can collectively enhance employment elasticity across the economy. Institutional mechanisms such as the National Care Economy Accelerator and India SkillPass can bring women and informal workers into formal and protected employment systems. Combined with universal social protection and portable digital benefits, these measures can strengthen labor mobility, reduce vulnerability, and provide long-term security.

Ensuring gender-sensitive infrastructure, including safe transport, affordable childcare, and equitable workplace norms, is critical to improving female labor force participation. Monitoring frameworks such as quarterly labor dashboards and a National Employment Council can institutionalize data-driven policy adjustments, ensuring adaptive governance and responsiveness to labor market changes.

Ultimately, sustainable growth will depend not only on the speed of economic expansion but also on the number of meaningful and secure jobs it creates, particularly for women and youth. By aligning growth with inclusion, skills development, and employment quality, India can transform its trajectory into one that is both productive and equitable, creating a resilient labor market by 2047.

ANNEXURE: Understanding the Quality of Growth Index

Economic growth alone does not guarantee broad-based prosperity, particularly if it fails to generate employment, improve job quality, or ensure inclusivity. The Quality of Growth (QoG) Index is designed to assess the nature of economic growth across Indian states, moving beyond traditional GDP measures.

This annexure details the methodology behind the QoG Index, including indicator selection, data normalization, composite scoring, and state rankings. It also explains how employment patterns, job quality, and structural shifts were incorporated to evaluate the effectiveness of economic growth at the state level. A robustness analysis was conducted to ensure the index provides a stable and reliable assessment of growth quality across different regions.

By quantifying the inclusivity and sustainability of growth, the QoG Index serves as a valuable tool for policymakers to identify strengths, address disparities, and design targeted interventions for employment generation and economic resilience.

Methodology Selection of Indicator

The first step in constructing the index was identifying suitable indicators. Given the availability of data in the public domain and the key performance indicators for this white paper, the following indicators were identified and utilized to calculate an index score. These indicators track the economic changes that occurred between 2017-18 and 2023-24 for total workforce, female workforce and youth workforce.

- Overall unemployment rate.
- Structural transformation from agriculture to non- agriculture sector.
- Employment type: to monitor self-employment, casual labor, regular wage or salaried jobs.
- Job quality: to monitor jobs without any Social Security Benefits.
- Per capita Income (PCI) growth rate over the past 5 years.

Data for the selected indicators was derived from PLFS data and national account statistics. Where State/Union Territory-level data was unavailable for any indicator, or for any population group, appropriate data was interpolated from the existing data series using relevant statistical methods and used to fill the missing gaps in the dataset. Data for the Union Territories Lakshadweep, Dadra & Nagar Haveli, Daman & Diu Islands was not included due to lack of complete data. Furthermore, the economic and demographic contributions of these regions are too insignificant to have a substantial effect on the index.

To analyse the impact of economic growth on the KPIs from 2017-18 to 2023-24, data was selected exclusively from two PLFS reports: 2017-18 and 2023-24.

Normalisation of raw indicator values: Normalization of raw indicator values to arrive at normalized scores on a standard scale of 0 to 1 was necessary to ensure comparability of values because different indicators had different ranges of values and units. Min-max normalization ensures all indicators are comparable on a uniform scale (0-1), preserving relative differences in performance while eliminating unit disparities.

For positive indicator series (where higher value means better performance), for example, change in share of employment in industries, the following formula was used:

$x' = x - min(x) / max(x) - min(x)$

Where, x = raw data value

min(x) = minimum observed value of the indicator in the dataset

max(x) = maximum value for the indicator

x' = normalised value after rescaling

For negative indicator series, (where higher value implies lower performance) for instance, Unemployment Rate, the following formula was used:

$x' = max(x) - x / max(x) - min(x)$

Where, x = raw data value

max(x) = maximum observed value of the indicator in the dataset

min(x) = minimum value for the indicator

x' = normalised value after rescaling

Aggregation of raw data into measurable indicator scores

Calculation of the score for the indicator 'PCI growth':

First, the arithmetic mean of the growth rate for the last 5 years was calculated for each state. Next, the mean value was normalized using min-max normalization to determine the score.

Calculation of the score for the indicator ‘overall unemployment score’:

First, the change in unemployment rates for the entire populations, females, and youth between the years 2017-18 to 2023-24 were calculated from the PLFS reports. Next, these values were individually normalized, and an arithmetic mean was calculated. This arithmetic mean of the normalized values was considered the score for the overall unemployment indicator.

Calculation of the score for the indicator 'Sectoral share in employment':

First, the change in the share of employment in agriculture, industry, and services was calculated separately for the entire population and for females between the years 2017-18 and 2023-24. Then, the values were normalized, and the arithmetic mean of the normalized values was used as the score for the indicator 'Sectoral share in employment.' Data for this indicator was not available at the state level for Youth; therefore, it was not included in the score calculation.

Calculation of the score for the indicator ‘Overall employment type’:

First, change in self-employment for the overall population and females, change in regular wage or salaried employment, and change in casual employment for the overall population and females were calculated separately and normalized between the years 2017-18 and 2023-24. The arithmetic mean of the normalized values was then used as the score for the ‘overall employment type’ indicator. Youth data for this indicator was not available at the state level and was therefore not included in the score calculation.

Calculation of the score for the indicator ‘Overall Job quality’:

First, the change in employment type with no social security benefits was calculated separately for the overall population and for females between the years 2017-18 and 2023-24. These values were then normalized, and the arithmetic mean of the normalized values was used as the score for the indicator 'Overall Job quality’'. As youth data for this indicator was not available at the state level, it was not included in the score calculation.

Computation of Composite Index Scores and Ranking the states:

The final step was the computation of composite Quality of Growth Index scores for the States/UTs. The composite score is the arithmetic mean of the normalised score of the five chosen indicators. All selected indicators were given equal weight to ensure a balanced assessment of economic growth. The rationale for equal weighting is that each indicator captures a distinct dimension of growth, and varying weights may introduce subjective bias. Finally, the States/UTs were then ranked according to the QoG scores.

Robustness Analysis:

To assess the reliability of the Quality of Growth (QoG) Index, a sensitivity test was conducted by randomly altering indicator weights and observing its effect on rankings. Further, an alternative index was computed using a weighted approach, assigning higher importance to job quality and employment type indicators. The results remained largely consistent with the original rankings, confirming that the QoG Index provides a robust and stable measure of growth quality across Indian states. The results showed an average rank deviation of 2.125 and an average score deviation of 0.0188, indicating minor fluctuations but no major distortions in rankings. Only Nagaland exhibited a substantial variation in rank during the sensitivity analysis. Given this validation, the index serves as a reliable tool for policymakers to assess economic performance beyond traditional GDP measures.

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